Architecture Justification — Customizable Subscription Box Scheduler

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Architecture Selection: We chose Microservices Architecture for the Customizable Subscription Box Scheduler.

Reason 1: Independently Scaling, Load-Diverse Concerns

The system has sharply different load profiles across functions: the Subscription Service handles continuous, low-latency subscriber traffic (browsing, customizing, pausing), while the Fulfillment & Inventory Service must burst to export shipping labels for 10,000 monthly boxes in under a minute (NFR-001). Microservices let each service scale on its own schedule and hardware profile instead of forcing one monolith to be provisioned for the fulfillment burst at all times.

Reason 2: Fault Isolation Across Distinct Actor Journeys

Subscribers and the Fulfillment Lead use the system for very different purposes, and the 48-hour swap/pause window (FR-001) has a strict 99.9% uptime requirement (NFR-002). By separating Subscription, Billing, Fulfillment, and Notification into independent services, an outage or slowdown in one — for example, the Notification Service during a bulk renewal-reminder send — does not take down the customer-facing customization portal that NFR-002 protects.

Security Advantage

Centralizing authentication in its own Auth Service means every other service (Subscription, Billing, Fulfillment, Notification) consumes identity through one well-defined IAuthenticate interface rather than each implementing its own login/session logic. This narrows where credentials and session tokens are handled, makes it easier to enforce TLS 1.2+ and encrypted transport consistently (NFR-001), and limits the blast radius if any single service is compromised.

Performance Benefit

Because the Fulfillment & Inventory Service is deployed and scaled independently, it can be provisioned with the batch-processing resources needed to generate manifests and export shipping labels for 10,000 boxes in under a minute (NFR-001), without over-provisioning the Subscription Service, which instead only needs to meet the lighter 3-second page-load target (NFR-002) for everyday browsing and customization traffic.

COMPONENT DIAGRAM